The Poisson Distribution

Section 1: The Poisson Probability Mass Function

1 If $X \sim \text{Po}(\lambda)$, what is the probability mass function $P(X = k)$ for $k = 0, 1, 2, \ldots$?

a. $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$

b. $P(X = k) = \frac{\lambda^k e^{\lambda}}{k!}$

c. $P(X = k) = \binom{n}{k}\lambda^k(1-\lambda)^{n-k}$

d. $P(X = k) = \lambda e^{-\lambda k}$

2 A call center receives an average of 3 calls per minute ($X \sim \text{Po}(3)$). What is the probability of receiving exactly 2 calls in a given minute?

a. $\frac{9}{2}e^{-3} \approx 0.224$

b. $3e^{-3}$

c. $\frac{27}{6}e^{-3}$

d. e^{-3}

3 For $X \sim \text{Po}(\lambda)$, what is $P(X = 0)$?

a. $e^{-\lambda}$

b. 0

c. 1

d. $\lambda e^{-\lambda}$

Section 2: Mean, Variance, and Poisson Characteristics

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4 What is a distinctive mathematical property of any Poisson random variable $X \sim \text{Po}(\lambda)$?

a. The mean and variance are strictly equal: $E(X) = \text{Var}(X) = \lambda$

b. The mean is always λ^2

c. The standard deviation equals λ

d. The median always equals $\lambda + 1$

5 If a Poisson random variable has variance $\text{Var}(X) = 16$, what is its standard deviation?

a. 4

b. 16

c. 256

d. $\sqrt{4} = 2$

6 Under what practical conditions does a Poisson process serve as a valid model?

a. Events occur independently, singly, and at a constant average rate across continuous time or space

b. There are exactly two trials with equal probability

c. Events must occur in clusters at predetermined times

d. The probability of an event increases as time elapses

Section 3: Changing Time Intervals & Sum of Independent Poisson RVs

7 Flaws on a wire occur at an average rate of 1.5 per meter. What is the distribution of flaws Y in a 4-meter segment of wire?

a. $Y \sim \text{Po}(6)$

b. $Y \sim \text{Po}(1.5)$

c. $Y \sim \text{Po}(4)$

d. $Y \sim \text{Po}(9)$

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8 If $X \sim \text{Po}(4)$ and $Y \sim \text{Po}(7)$ are independent random variables, what is the distribution of $W = X + Y$?

a. $W \sim \text{Po}(11)$

b. $W \sim \text{Po}(28)$

c. $W \sim \text{Po}(3)$

d. $W \sim N(11, 28)$

9 If $W \sim \text{Po}(11)$, what is the variance of W?

a. 11

b. 121

c. $\sqrt{11}$

d. 22

Section 4: Poisson Approximation to the Binomial Distribution

10 When can a Binomial distribution $B(n, p)$ be well approximated by a Poisson distribution $\text{Po}(\lambda)$?

a. When n is very large ($n > 50$) and p is very small ($p < 0.1$), with $\lambda = np$

b. When n is small and $p = 0.5$

c. When $np \geq 10$ and $n(1 - p) \geq 10$

d. For any binomial distribution where $n > 30$

11 A factory produces microchips with defect probability $p = 0.002$. In a batch of $n = 1000$ chips, approximate the probability of finding zero defects.

a. $e^{-2} \approx 0.135$

b. $e^{-0.002}$

c. $1 - e^{-2}$

d. 0.002

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12 If $X \sim \text{Po}(\lambda)$ has $P(X = 1) = P(X = 2)$, find the value of λ.

a. $\lambda = 2$

b. $\lambda = 1$

c. $\lambda = 0.5$

d. $\lambda = 4$

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